

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)

DEEP LEARNING LAB RECORD

LAB - 04

Student Details

Name: P. Harshith

Register Number: CH.SC.U4AIE24057

Course Name: Deep Learning

Course Code: 22AIE304

Experiment Title

Deep Neural Network (DNN) for Handwritten Digit Classification using the MNIST Dataset

Aim

To build and train a Deep Neural Network (DNN) using TensorFlow/Keras for handwritten digit classification on the MNIST dataset.

Objective

- Load the MNIST dataset.
 - Preprocess image data.
 - Build a Deep Neural Network.
 - Train the model.
 - Evaluate its performance.
 - Predict handwritten digits.
-

Dataset

Dataset Name: MNIST Handwritten Digits

Training Images: 60,000

Testing Images: 10,000

Image Size: 28 × 28 pixels

Classes: 0–9

Software Requirements

- Python 3.x
 - TensorFlow
 - NumPy
 - Matplotlib
 - Scikit-Learn
 - Jupyter Notebook
-

Required Libraries

- tensorflow
- numpy
- matplotlib

```
In [1]: !pip install tensorflow matplotlib numpy
```

Requirement already satisfied: tensorflow in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.21.0)

Requirement already satisfied: matplotlib in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (3.10.9)

Requirement already satisfied: numpy in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.2.6)

Requirement already satisfied: google_pasta>=0.1.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.2.0)

Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (25.12.19)

Requirement already satisfied: keras>=3.12.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.12.3)

Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.4.0)

Requirement already satisfied: wrapt>=1.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.2.2)

Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.14.0)

Requirement already satisfied: six>=1.12.0 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (4.15.0)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (1.81.1)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: packaging in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (25.0)

Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (7.35.0)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.7.0)

Requirement already satisfied: setuptools in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (65.5.0)

Requirement already satisfied: libclang>=13.0.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (18.1.1)

Requirement already satisfied: requests<3,>=2.21.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.34.2)

Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.5.4)

Requirement already satisfied: termcolor>=1.1.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.3.0)

Requirement already satisfied: absl-py>=1.0.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.4.0)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\msi\appdata\roaming\python\python310\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: pillow>=8 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (12.2.0)

Requirement already satisfied: pyparsing>=3 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (3.3.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.3.2)

Requirement already satisfied: cycler>=0.10 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (4.63.0)

Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\msi\appdata\local\p

```
programs\python\python310\lib\site-packages (from astunparse>=1.6.0->tensorflow)
(0.47.0)
Requirement already satisfied: rich in c:\users\msi\appdata\local\programs\python
\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (15.0.0)
Requirement already satisfied: optree in c:\users\msi\appdata\local\programs\pyth
on\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (0.19.1)
Requirement already satisfied: namex in c:\users\msi\appdata\local\programs\pytho
n\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\msi\appdata\local\pro
grams\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2.7.0)
Requirement already satisfied: certifi>=2023.5.7 in c:\users\msi\appdata\local\pr
ograms\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2026.5.20)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\msi\appdata\l
ocal\programs\python\python310\lib\site-packages (from requests<3,>=2.21.0->tenso
rflow) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in c:\users\msi\appdata\local\program
s\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.1
6)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\msi\appdata\loca
l\programs\python\python310\lib\site-packages (from rich->keras>=3.12.0->tensorfl
ow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\msi\appdata\ro
aming\python\python310\site-packages (from rich->keras>=3.12.0->tensorflow) (2.1
9.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\msi\appdata\local\programs
\python\python310\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.1
2.0->tensorflow) (0.1.2)

[notice] A new release of pip is available: 23.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
In [2]: !pip install tensorflow matplotlib numpy
```

Requirement already satisfied: tensorflow in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.21.0)

Requirement already satisfied: matplotlib in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (3.10.9)

Requirement already satisfied: numpy in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.2.6)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (1.81.1)

Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.7.0)

Requirement already satisfied: google_pasta>=0.1.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.2.0)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: protobuf<8.0.0,>=6.31.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (7.35.0)

Requirement already satisfied: six>=1.12.0 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: libclang>=13.0.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (18.1.1)

Requirement already satisfied: keras>=3.12.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.12.3)

Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (25.12.19)

Requirement already satisfied: wrapt>=1.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.2.2)

Requirement already satisfied: opt_einsum>=2.3.2 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.4.0)

Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (4.15.0)

Requirement already satisfied: packaging in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (25.0)

Requirement already satisfied: termcolor>=1.1.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.3.0)

Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.14.0)

Requirement already satisfied: setuptools in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (65.5.0)

Requirement already satisfied: requests<3,>=2.21.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.34.2)

Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.5.4)

Requirement already satisfied: absl-py>=1.0.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.4.0)

Requirement already satisfied: pillow>=8 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (12.2.0)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\msi\appdata\roaming\python\python310\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: pyparsing>=3 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (3.3.2)

Requirement already satisfied: cycycler>=0.10 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (4.63.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.3.2)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\msi\appdata\local\p

```

programs\python\python310\lib\site-packages (from astunparse>=1.6.0->tensorflow)
(0.47.0)
Requirement already satisfied: optree in c:\users\msi\appdata\local\programs\pyth
on\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (0.19.1)
Requirement already satisfied: namex in c:\users\msi\appdata\local\programs\pytho
n\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: rich in c:\users\msi\appdata\local\programs\python
\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (15.0.0)
Requirement already satisfied: certifi>=2023.5.7 in c:\users\msi\appdata\local\pr
ograms\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2026.5.20)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\msi\appdata\local\pro
grams\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2.7.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\msi\appdata\l
ocal\programs\python\python310\lib\site-packages (from requests<3,>=2.21.0->tenso
rflow) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in c:\users\msi\appdata\local\program
s\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.1
6)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\msi\appdata\loca
l\programs\python\python310\lib\site-packages (from rich->keras>=3.12.0->tensorfl
ow) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\msi\appdata\ro
aming\python\python310\site-packages (from rich->keras>=3.12.0->tensorflow) (2.1
9.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\msi\appdata\local\programs
\python\python310\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.1
2.0->tensorflow) (0.1.2)
[notice] A new release of pip is available: 23.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

```

```

In [3]: import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

```

```

In [4]: # Load MNIST Dataset

(X_train, y_train), (X_test, y_test) = tf.keras.datasets.mnist.load_data()

print(X_train.shape)
print(X_test.shape)

```

```

Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-dataset
s/mnist.npz
11490434/11490434 ————— 5s 0us/step
(60000, 28, 28)
(10000, 28, 28)

```

```

In [5]: # Display Sample Images

plt.figure(figsize=(10,5))

for i in range(10):

    plt.subplot(2,5,i+1)

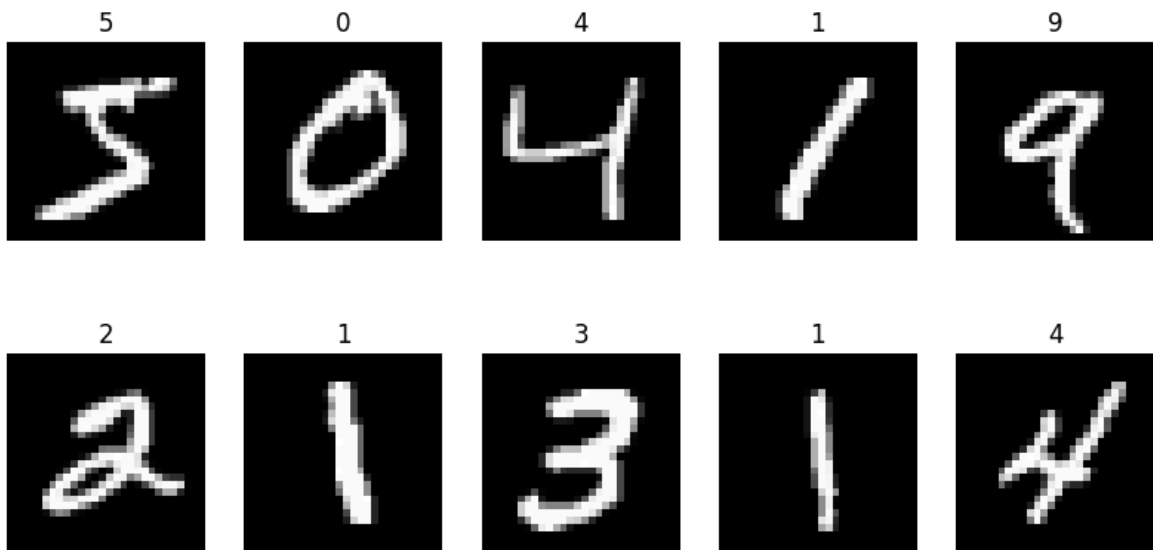
```

```
plt.imshow(X_train[i], cmap="gray")

plt.title(y_train[i])

plt.axis("off")

plt.show()
```



```
In [6]: # Normalize Images

X_train = X_train / 255.0

X_test = X_test / 255.0
```

```
In [7]: # Build Deep Neural Network

model = Sequential()

model.add(Flatten(input_shape=(28,28)))

model.add(Dense(128, activation='relu'))

model.add(Dense(64, activation='relu'))

model.add(Dense(10, activation='softmax'))

model.summary()
```

c:\Users\msi\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\reshaping\flatten.py:37: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(**kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None , 784)	0
dense (Dense)	(None , 128)	100,480
dense_1 (Dense)	(None , 64)	8,256
dense_2 (Dense)	(None , 10)	650



Total params: [109,386](#) (427.29 KB)

Trainable params: [109,386](#) (427.29 KB)

Non-trainable params: [0](#) (0.00 B)

In [8]: *# Compile Model*

```
model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

In [9]: *# Train Model*

```
history = model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32,
    validation_split=0.2
)
```


Epoch 1/10
1500/1500 ————— 3s 2ms/step - accuracy: 0.9221 - loss: 0.2693 - val_accuracy: 0.9515 - val_loss: 0.1580
Epoch 2/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9646 - loss: 0.1172 - val_accuracy: 0.9584 - val_loss: 0.1352
Epoch 3/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9753 - loss: 0.0799 - val_accuracy: 0.9715 - val_loss: 0.0973
Epoch 4/10
1500/1500 ————— 2s 2ms/step - accuracy: 0.9805 - loss: 0.0603 - val_accuracy: 0.9693 - val_loss: 0.1054
Epoch 5/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9852 - loss: 0.0467 - val_accuracy: 0.9739 - val_loss: 0.0912
Epoch 6/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9886 - loss: 0.0359 - val_accuracy: 0.9718 - val_loss: 0.1061
Epoch 7/10
1500/1500 ————— 2s 2ms/step - accuracy: 0.9904 - loss: 0.0297 - val_accuracy: 0.9708 - val_loss: 0.1027
Epoch 8/10
1500/1500 ————— 2s 2ms/step - accuracy: 0.9908 - loss: 0.0278 - val_accuracy: 0.9758 - val_loss: 0.0994
Epoch 9/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9926 - loss: 0.0226 - val_accuracy: 0.9727 - val_loss: 0.1223
Epoch 10/10
1500/1500 ————— 2s 1ms/step - accuracy: 0.9940 - loss: 0.0177 - val_accuracy: 0.9739 - val_loss: 0.1170

In [11]: *# Evaluate Model*

```
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Loss :", loss)

print("Test Accuracy :", accuracy)
```

313/313 ————— 0s 917us/step - accuracy: 0.9799 - loss: 0.0932
Test Loss : 0.0931696817278862
Test Accuracy : 0.9799000024795532

In [12]: *# Predict Digits*

```
predictions = model.predict(X_test)

predicted_labels = np.argmax(predictions, axis=1)

print(predicted_labels[:10])
```

313/313 ————— 0s 759us/step
[7 2 1 0 4 1 4 9 5 9]

In [13]: *# Compare Actual and Predicted*

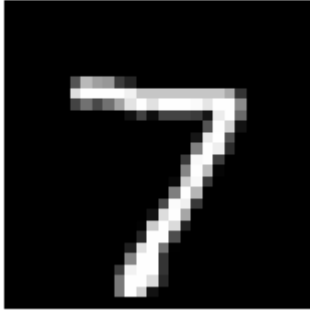
```
for i in range(5):

    plt.figure(figsize=(2,2))

    plt.imshow(X_test[i], cmap='gray')
```

```
plt.title(f"Actual : {y_test[i]} Predicted : {predicted_labels[i]}")  
plt.axis('off')  
plt.show()
```

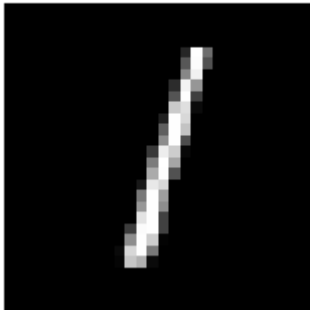
Actual : 7 Predicted : 7



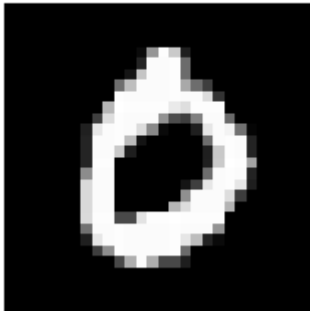
Actual : 2 Predicted : 2



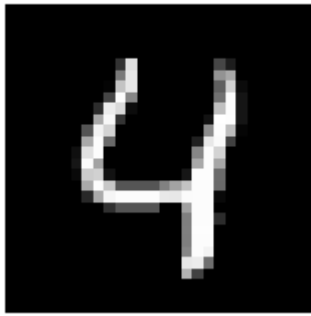
Actual : 1 Predicted : 1



Actual : 0 Predicted : 0

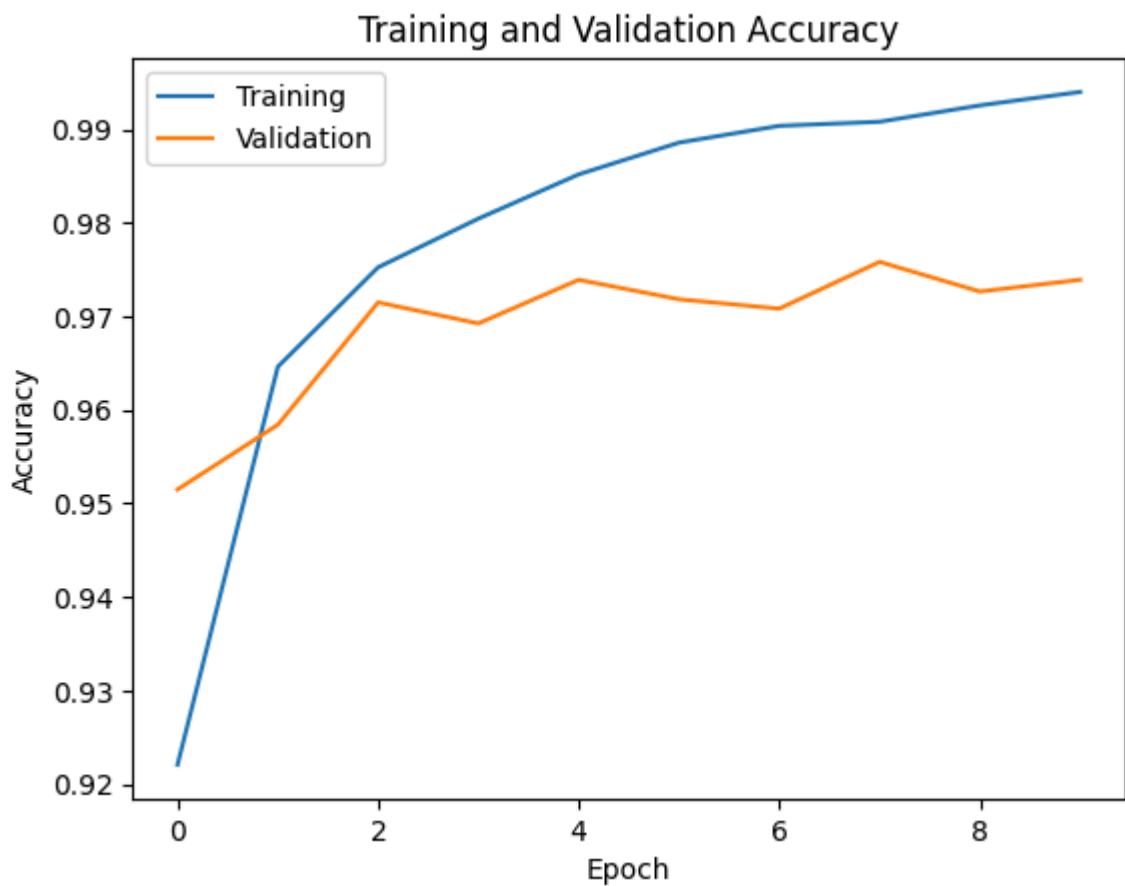


Actual : 4 Predicted : 4



In [14]: *# Plot Accuracy*

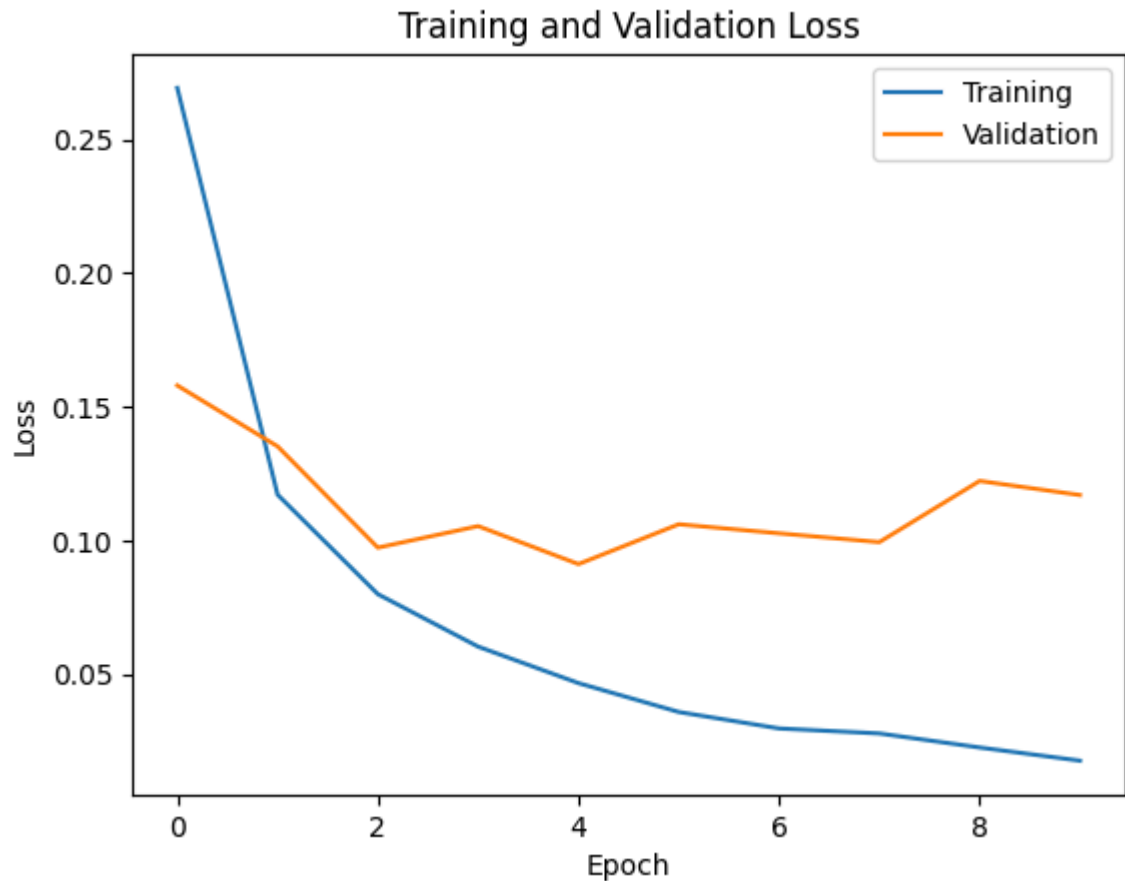
```
plt.plot(history.history['accuracy'])  
plt.plot(history.history['val_accuracy'])  
plt.title("Training and Validation Accuracy")  
plt.xlabel("Epoch")  
plt.ylabel("Accuracy")  
plt.legend(["Training", "Validation"])  
plt.show()
```



In [15]: *# Plot Loss*

```
plt.plot(history.history['loss'])
```

```
plt.plot(history.history['val_loss'])  
  
plt.title("Training and Validation Loss")  
  
plt.xlabel("Epoch")  
  
plt.ylabel("Loss")  
  
plt.legend(["Training", "Validation"])  
  
plt.show()
```



```
In [16]: # Classification Report  
  
from sklearn.metrics import classification_report  
  
print(classification_report(y_test, predicted_labels))
```

	precision	recall	f1-score	support
0	0.97	0.99	0.98	980
1	0.99	0.99	0.99	1135
2	0.99	0.96	0.97	1032
3	0.97	0.99	0.98	1010
4	0.98	0.98	0.98	982
5	0.99	0.98	0.98	892
6	0.99	0.98	0.98	958
7	0.98	0.98	0.98	1028
8	0.97	0.98	0.97	974
9	0.98	0.97	0.97	1009
accuracy			0.98	10000
macro avg	0.98	0.98	0.98	10000
weighted avg	0.98	0.98	0.98	10000

In [17]: *# Confusion Matrix*

```

from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, predicted_labels)

print(cm)

```

```

[[ 974    0    0    0    0    1    2    1    2    0]
 [   0 1125    3    2    0    0    0    2    2    1]
 [   7    0  993    7    4    0    5    9    7    0]
 [   1    0    1  998    0    2    0    3    4    1]
 [   2    0    3    1  965    0    2    1    2    6]
 [   2    0    0    8    1  870    2    1    5    3]
 [   7    4    0    1    2    3  940    0    1    0]
 [   1    4    3    2    0    0    1 1008    2    7]
 [   3    0    3    5    2    2    1    3  951    4]
 [   2    2    0    9   10    4    0    5    2  975]]

```